

**2021-2022 Fall**  
**MAT123 Final**  
**07/01/2022**

1.
  - a. Find  $\int \frac{dx}{x^3 - 4x^2 + 3x}$ .
  - b. Evaluate  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\pi/2}^x \ln(\sin t) dt}{\sin x - 1}$ .
  - c. Evaluate the improper integral  $\int_1^2 \frac{dx}{(x-1)^{2/3}}$ .
2. Consider the region  $R$  bounded by the curves  $y = \arctan x$ ,  $y = \ln x$  and the lines  $x = \frac{1}{\sqrt{3}}$  and  $x = 1$ .
  - a. Find the area of the region  $R$ .
  - b. Write a definite integral (do not evaluate) by using the Washer Method, which gives the volume of a solid obtained by rotating the region  $R$  about the  $y$ -axis.
  - c. Write a definite integral (do not evaluate) by using the Cylindrical Shell Method, which gives the volume of a solid obtained by rotating the region  $R$  about the line  $y = 2$ .
3. Determine whether each series is convergent or divergent. Explain your answer.
  - a.  $\sum_{n=1}^{\infty} (\arctan n - \arctan(n-1))$
  - b.  $\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}}$
  - c.  $\sum_{n=1}^{\infty} \cos\left(\frac{n^2}{1+n^4}\right)$
  - d.  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

1. a. Use the method of partial fraction decomposition.

$$\int \frac{dx}{x^3 - 4x^2 + 3x} = \int \frac{dx}{x(x-3)(x-1)} = \int \left( \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x-1} \right) dx$$

$$\begin{aligned} A(x-3)(x-1) + Bx(x-1) + Cx(x-3) &= 1 \\ x^2(A+B+C) + x(-4A-B-3C) + 3A &= 1 \end{aligned}$$

Equate the coefficients of like terms.

$$\left. \begin{aligned} x^2(A+B+C) &= 0 \\ x(-4A-B-3C) &= 0 \\ 3A &= 1 \end{aligned} \right\} \rightarrow A = \frac{1}{3}, \quad \left. \begin{aligned} B+C &= -\frac{1}{3} \\ B+3C &= -\frac{4}{3} \end{aligned} \right\} \rightarrow B = \frac{1}{6}, \quad C = -\frac{1}{2}$$

Rewrite the integral.

$$\begin{aligned} \int \left( \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x-1} \right) dx &= \int \left( \frac{1}{3x} + \frac{1}{6(x-3)} - \frac{1}{2(x-1)} \right) dx \\ &= \boxed{\frac{1}{3} \ln |x| + \frac{1}{6} \ln |x-3| - \frac{1}{2} \ln |x-1| + c, \quad c \in \mathbb{R}} \end{aligned}$$

- b. The limit is in the indeterminate form  $0/0$ . Apply L'Hôpital's rule to eliminate the indeterminate form.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\pi/2}^x \ln(\sin t) dt}{\sin x - 1} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{d}{dx} \int_{\pi/2}^x \ln(\sin t) dt}{\cos x}$$

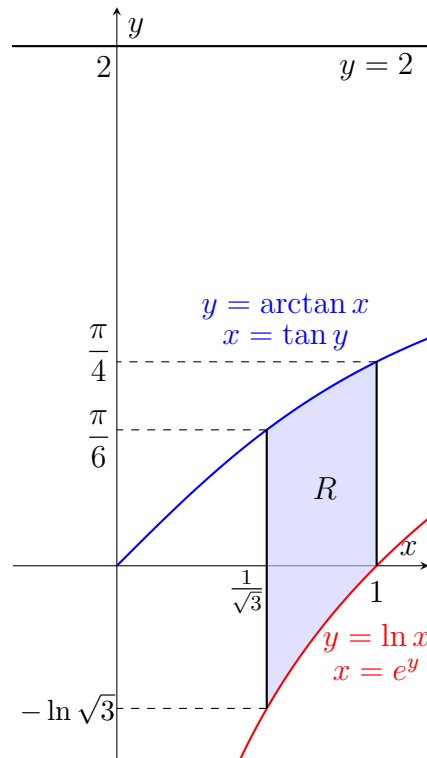
By the Fundamental Theorem of Calculus, we may rewrite the limit as follows.

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{d}{dx} \int_{\pi/2}^x \ln(\sin t) dt}{\cos x} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{\cos x} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sin x} \cdot \cos x}{-\sin x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{-\sin^2 x} = -\frac{\cos \frac{\pi}{2}}{\sin^2 \frac{\pi}{2}} = \boxed{0} \end{aligned}$$

- c. Take the limit since this is an improper integral.

$$\begin{aligned} \int_1^2 \frac{dx}{(x-1)^{2/3}} &= \lim_{R \rightarrow 1^+} \int_R^2 \frac{dx}{(x-1)^{2/3}} = \lim_{R \rightarrow 1^+} 3(x-1)^{1/3} \Big|_R^2 \\ &= 3 \lim_{R \rightarrow 1^+} (1 - (R-1)^{1/3}) = \boxed{3} \end{aligned}$$

2.



a. Set up the integral.

$$A = \int_{1/\sqrt{3}}^1 (\arctan x - \ln x) dx = \int_{1/\sqrt{3}}^1 \arctan x dx - \int_{1/\sqrt{3}}^1 \ln x dx \quad (1)$$

Calculate the first integral in (1) by integration by parts.

$$\left. \begin{aligned} u = \arctan x &\implies du = \frac{1}{x^2 + 1} dx \\ dv = dx &\implies v = x \end{aligned} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\begin{aligned} \int_{1/\sqrt{3}}^1 \arctan x dx &= x \arctan x \Big|_{1/\sqrt{3}}^1 - \int_{1/\sqrt{3}}^1 \frac{x}{x^2 + 1} dx \\ &= \left( x \arctan x - \frac{1}{2} \ln |x^2 + 1| \right) \Big|_{1/\sqrt{3}}^1 \\ &= \left( \frac{\pi}{4} - \frac{\ln 2}{2} \right) - \left( \frac{\pi\sqrt{3}}{18} - \frac{1}{2} \ln \left( \frac{4}{3} \right) \right) = \frac{\pi(9 - 2\sqrt{3})}{36} + \frac{1}{2} \ln \frac{2}{3} \end{aligned}$$

Calculate the second integral in(1) by integration by parts.

$$\left. \begin{array}{l} u = \ln x \implies du = \frac{1}{x} dx \\ dv = dx \implies v = x \end{array} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\begin{aligned} \int_{1/\sqrt{3}}^1 \ln x dx &= x \ln x \Big|_{1/\sqrt{3}}^1 - \int_{1/\sqrt{3}}^1 dx = (x \ln x - x) \Big|_{1/\sqrt{3}}^1 \\ &= (0 - 1) - \left( -\frac{\ln \sqrt{3}}{\sqrt{3}} - \frac{1}{\sqrt{3}} \right) = \frac{\sqrt{3} \ln \sqrt{3} + \sqrt{3} - 3}{3} \end{aligned}$$

The result is then

$$A = \frac{\pi (9 - 2\sqrt{3})}{36} + \frac{1}{2} \cdot \ln \frac{2}{3} - \frac{\sqrt{3} \ln \sqrt{3} + \sqrt{3} - 3}{3}$$

b.

$$\begin{aligned} &\int_{-\ln \sqrt{3}}^0 \pi \left[ (e^y)^2 - \left( \frac{1}{\sqrt{3}} \right)^2 \right] dy + \int_0^{\pi/6} \pi \left[ (1)^2 - \left( \frac{1}{\sqrt{3}} \right)^2 \right] dy \\ &+ \int_{\pi/6}^{\pi/4} \pi \left[ (1^2) - (\tan y)^2 \right] dy \end{aligned}$$

c.

$$\begin{aligned} &\int_{-\ln \sqrt{3}}^0 2\pi (2 - y) \left( e^y - \frac{1}{\sqrt{3}} \right) dy + \int_0^{\pi/6} 2\pi (2 - y) \left( 1 - \frac{1}{\sqrt{3}} \right) dy \\ &+ \int_{\pi/6}^{\pi/4} 2\pi (2 - y) (1 - \tan y) dy \end{aligned}$$

3. a. Determine the  $n$ th partial sum.

$$\begin{aligned} \sum_{n=1}^{\infty} (\arctan n - \arctan(n-1)) &= (\cancel{\arctan 1} - \arctan 0) + (\cancel{\arctan 2} - \cancel{\arctan 1}) \\ &+ (\cancel{\arctan 3} - \cancel{\arctan 2}) + (\cancel{\arctan 4} - \cancel{\arctan 3}) \\ &+ \cdots + (\arctan n - \cancel{\arctan(n-1)}) \\ &= \arctan n - \arctan 0 = \arctan n \end{aligned}$$

$$\lim_{n \rightarrow \infty} \arctan n = \frac{\pi}{2} \quad (\text{convergent})$$

Therefore, the series  $\sum_{n=1}^{\infty} (\arctan n - \arctan(n-1))$  converges.

b. Rewrite the sum.

$$\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}} = \sum_{n=0}^{\infty} \frac{-3 \cdot (-3)^n}{8^n} = -3 \sum_{n=0}^{\infty} \left(-\frac{3}{8}\right)^n$$

This is a geometric series where  $r = -\frac{3}{8}$ .  $|r| = \frac{3}{8} < 1$ . Therefore, the series

$$\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}} \text{ converges.}$$

c. Apply the  $n$ th Term Test for divergence. We may take the limit inside the trigonometric function because  $\cos x$  is continuous everywhere. Since  $1 + n^4$  grows faster than  $n^2$ , the value of the expression  $\frac{n^2}{1 + n^4}$  tends to zero.

$$\lim_{n \rightarrow \infty} \cos\left(\frac{n^2}{1 + n^4}\right) = \cos\left[\lim_{n \rightarrow \infty} \left(\frac{n^2}{1 + n^4}\right)\right] = \cos 0 = 1 \neq 0$$

By the  $n$ th Term Test for divergence, the series  $\sum_{n=1}^{\infty} \cos\left(\frac{n^2}{1 + n^4}\right)$  diverges.

d. Take  $f(x) = \frac{1}{x(\ln x)^2}$ .  $f$  is positive and decreasing for  $x \geq 2$  because  $x$  and  $(\ln x)^2$  are positive and increasing for  $x \geq 2$ .  $x$  is a polynomial which is defined everywhere and  $(\ln x)^2$  is continuous for  $x \geq 2$ . Since we took into account every criterion, we may apply the Integral Test. Handle the improper integrals by taking the limit.

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \lim_{R \rightarrow \infty} \int_2^R \frac{dx}{x(\ln x)^2} = \lim_{R \rightarrow \infty} \left[-\frac{1}{\ln x}\right]_2^R = \lim_{R \rightarrow \infty} \left[-\frac{1}{\ln R} + \frac{1}{\ln 2}\right] = \frac{1}{\ln 2}$$

Since the integral converges, the series  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$  also converges.